

GlowCart Beauty

Sales & Profitability Analysis Report

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This report presents a comprehensive end-to-end data analysis of GlowCart Beauty's 2023 sales operations. It covers data engineering, exploratory data analysis, SQL-driven business queries, actionable insights, and strategic recommendations — all grounded in 5,000 transaction records across 5 regions, 5 sales channels, 20 products, and 15 sales representatives.

01 | Business Context & Problem Statement

GlowCart Beauty, a multi-channel beauty products company, was experiencing a critical challenge: despite growing transaction volumes, leadership lacked visibility into where profit was actually being made — and more importantly, where it was silently eroding. Discounts were being offered without a clear profitability framework, regional performance disparities were unquantified, and sales-rep contribution to margin (not just revenue) was unknown.

***The primary business problem:** “Which combinations of region, channel, product, and sales rep drive true profitability — and where is the business leaking margin through uncontrolled discounting?”*

Without answers to this, the business risked over-investing in high-revenue but low-margin segments, under-rewarding high-performing reps, and continuing a discounting policy that was quietly destroying █14+ million in potential profit annually.

02 | Analytical Task & Technical Approach

To solve this, a full-stack data analysis pipeline was built, spanning raw data ingestion to SQL-based business intelligence queries and Power BI visualization.

Technical Stack & Pipeline

Stage	Tool / Technique	Purpose
Data Ingestion	Python (pandas)	Load raw Excel → DataFrame
Data Cleaning	pandas .dropna(), .duplicated()	Handle nulls, validate schema
Date Engineering	pd.to_datetime(), .dt accessor	Extract Year, Month, Quarter
Feature Engineering	Derived column formulas	Gross Revenue, Discount Amount, Net Revenue, COGS, Gross Profit, Margin %
Data Export	df.to_csv() + SQLAlchemy	Save cleaned CSV, push to MySQL
SQL Analysis	MySQL (19 queries)	Aggregations, rankings, cross-tab analysis
Visualization	Power BI (.pbix)	Interactive dashboards

Feature Engineering — Derived Columns

Six business-critical financial metrics were engineered from raw transactional data using arithmetic feature construction:

Feature	Formula	Business Meaning
Gross Revenue	Qty Sold x Unit Price	Top-line revenue before discounts
Discount Amount	Gross Revenue x (Discount% / 100)	Revenue surrendered via promotions
Net Revenue	Gross Revenue - Discount Amount	Actual revenue realized
Total COGS	Qty Sold x COGS Per Unit	Direct cost of goods sold
Gross Profit	Net Revenue - Total COGS	Core profitability metric
Gross Margin %	(Gross Profit / Net Revenue) x 100	Efficiency of each rupee of revenue

SQL Query Coverage (19 Business Queries)

After cleaning and feature engineering, the dataset was pushed to a MySQL database (glowcart_db) via SQLAlchemy. Nineteen structured SQL queries were written covering the following analytical dimensions:

- **Volume Metrics:** Total transactions, orders by region
- **Revenue Analysis:** Total revenue, revenue by region / month / quarter
- **Profitability:** Total profit, profit by region, top 10 profit products, least profitable products, average margin %
- **Channel Intelligence:** Revenue and profit by sales channel, top region+channel combos
- **Product Rankings:** Top 10 revenue products, top 10 profit products, products with highest discounts
- **Sales Rep Performance:** Top reps by revenue, top reps by profit
- **Discount Leakage:** Average discount %, total discount loss, products with highest avg discount

03 | Key Business Metrics — At a Glance

5,000 Total Transactions	Rs. 8.62 Cr Gross Revenue	Rs. 7.19 Cr Net Revenue
Rs. 3.89 Cr Total Gross Profit	55.94% Avg Gross Margin %	Rs. 1.43 Cr Total Discount Leakage

*Dataset period: Full Year 2023. Currency figures in Indian Rupees.

04 | Key Findings & Analytical Results

4.1 Regional Performance

North and Central regions lead both revenue and profit, but the margin story is more revealing — Central outperforms on profitability efficiency despite similar revenue to North.

Region	Net Revenue (Rs.)	Gross Profit (Rs.)	Avg Margin %
Central	1,58,96,658	91,93,514	60.05%
North	1,59,48,335	91,44,045	59.73%
East	1,52,41,073	87,47,232	59.73%
South	1,36,17,904	69,45,918	54.00%
West	1,11,91,379	48,52,322	45.79%

Critical Finding: West region's margin is 45.79% — a 14-point gap below Central (60.05%). This is not a revenue problem; it is a cost and discount-control problem in the West.

4.2 Sales Channel Performance

All five channels are nearly revenue-equivalent (Rs. 1.37–1.47 Cr each), indicating healthy diversification. However, Pharmacy channel leads on margin efficiency at 56.45%, making it the highest-quality revenue channel.

Channel	Net Revenue (Rs.)	Gross Profit (Rs.)	Avg Margin %
Pharmacy	1,46,19,548	80,03,377	56.45%
Salon Wholesale	1,39,56,591	76,05,199	56.31%
Online	1,46,66,695	79,38,534	55.79%
Modern Trade	1,45,56,457	77,79,231	55.65%
General Trade	1,40,96,059	75,56,691	55.51%

4.3 Product-Level Intelligence

Top Revenue Generator: Perfume 100ml EDP dominates with Rs. 1.21 Cr in net revenue — nearly 2x its nearest competitor (Night Repair Cream at Rs. 64 Lac). It also leads profit at Rs. 54.6 Lac (14% of total company profit).

Rank	Product	Net Revenue (Rs.)	Gross Profit (Rs.)
1	Perfume 100ml EDP	1,20,97,386	54,56,264
2	Night Repair Cream 50ml	64,04,011	35,64,597
3	Eyeshadow Palette 9-shade	48,45,826	22,82,409
4	Under Eye Cream 15ml	47,16,820	28,03,159
5	Blush Palette	46,49,947	22,37,582
6	Vitamin C Face Serum 30ml	42,20,660	26,23,089
7	BB Cream 30ml	36,26,131	19,89,577
8	Nail Polish Set (12pc)	35,33,614	17,25,124
9	Hair Serum 100ml	34,30,073	18,67,065
10	Mascara	30,47,189	15,30,000+

Least Profitable Products: Kajal Eye Pencil (Rs. 7.03 Lac), Lip Gloss (Rs. 10.54 Lac), and Hair Oil 200ml (Rs. 11.51 Lac) sit at the bottom — these are candidates for portfolio review or pricing revision.

4.4 Discount Leakage Analysis

This is one of the most financially impactful findings in the analysis.

16.53% Avg Discount Given	Rs. 1.43 Cr Total Revenue Lost	16.56% % of Gross Revenue
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Rs. 1.43 Crore — 16.56% of all gross revenue — was surrendered through discounts in 2023. The highest discounted products are Rose Water Toner 200ml (17.71% avg discount), Night Repair Cream 50ml (17.30%), BB Cream 30ml (17.14%), and Mascara (17.14%). Notably, Night Repair Cream and BB Cream are also among the top revenue products — meaning the business is over-discounting its best-sellers unnecessarily.

4.5 A/B Testing — Discount Impact Validation

To move beyond descriptive trends and statistically confirm whether discount depth causes a real difference in business outcomes, an independent-samples **Welch's t-test** was run in Python (*scipy.stats.ttest_ind*, *equal_var=False*) comparing low-discount transactions (Group A, discount ≤ 10%) against high-discount transactions (Group B, discount > 20%) across four metrics: Net Revenue, Gross Margin %, Quantity Sold, and Gross Profit. **Cohen's d** was then calculated for each metric to measure the practical size of each effect, not just its statistical significance (d < 0.2 = Negligible, 0.2–0.49 = Small, 0.5–0.79 = Medium, ≥0.8 = Large).

Metric	Mean (Group A)	Mean (Group B)	t-stat	p-value	Sig. (p < 0.05)	Cohen's d	Effect Size
Net Revenue	Rs. 15,782.85	Rs. 12,089.01	7.841	0.00000	Yes	0.268	Small
Gross Margin %	60.81%	48.44%	42.717	0.00000	Yes	1.501	Large
Qty Sold	30.53	30.88	-0.590	0.55503	No	-0.020	Negligible
Gross Profit	Rs. 9,223.26	Rs. 5,540.14	16.057	0.00000	Yes	0.545	Medium

Insight: High-discount transactions (Group B) show a statistically significant drop in net revenue, gross margin, and gross profit compared to low-discount transactions ($p < 0.00001$ on all three), and the **gross margin effect is not just significant but large in practical terms** (Cohen's $d = 1.501$) — nearly double the 0.8 threshold for a “large” effect. Gross profit shows a medium effect ($d = 0.545$) and net revenue a small one ($d = 0.268$), indicating the discount is eroding margin far more aggressively than it's eroding top-line revenue. Quantity sold, by contrast, shows no statistically significant difference between groups ($p = 0.555$) and a negligible effect size ($d = -0.020$) — meaning bigger discounts are **not** meaningfully driving higher unit sales. This disproves the common assumption that “discounting drives volume”: in this dataset, deep discounts erode revenue, profit, and — most severely — margin, without a corresponding lift in units sold, directly reinforcing the case for tighter discount guardrails (see R1).

4.6 Sales Representative Analysis

An important distinction was uncovered between reps who drive revenue vs. those who drive profit — these are not always the same person.

Sales Rep	Net Revenue (Rs.)	Gross Profit (Rs.)	Profit Rank
Meena G	58,95,950	32,92,749	#3
Vikram N	58,53,934	34,10,422	#1
Deepa S	56,73,540	33,27,179	#2
Anjali M	53,49,819	30,14,569	#5
Suresh K	53,25,100	30,28,095	#4

Key Insight: Meena G leads in revenue but ranks 3rd in profit. Vikram N generates slightly less revenue yet delivers the highest profit — indicating better discount discipline and product-mix management.

4.7 Temporal Trends

Q3 was the strongest quarter (Rs. 1.86 Cr), followed closely by Q4 (Rs. 1.81 Cr). March and September were the peak individual months at Rs. 65.3 Lac and Rs. 65.6 Lac respectively. June recorded the lowest monthly revenue at Rs. 51.7 Lac, suggesting a mid-year demand dip that could be addressed with targeted promotions.

Quarter	Net Revenue (Rs.)	% of Annual
Q1 (Jan–Mar)	1,76,33,290	24.5%
Q2 (Apr–Jun)	1,76,15,203	24.5%
Q3 (Jul–Sep)	1,85,61,210	25.8%
Q4 (Oct–Dec)	1,80,85,646	25.2%

05 | Business Impact of This Analysis

Before this analysis, GlowCart Beauty had raw transaction data but no structured view of profitability drivers. The analytical work delivered the following measurable impact:

Margin Visibility

The business now knows that West region's 45.79% margin is dragging company-wide profitability. A 5-point margin improvement in West alone would add ~Rs. 24 Lac to annual profit.

Discount Quantification

Rs. 1.43 Crore of discount leakage was identified and quantified for the first time. Leadership can now set data-backed discount thresholds per product.

Top SKU Concentration Risk

One product (Perfume 100ml EDP) generates 14% of total company profit. This portfolio concentration risk was invisible before the analysis.

Rep Performance Clarity

The gap between revenue rank and profit rank for reps was exposed. Incentive structures can now be shifted from pure-revenue to margin-adjusted metrics.

Seasonal Planning

A Rs. 14 Lac revenue gap between the best month (September) and worst (June) was quantified, enabling targeted campaign planning.

Best-Performing Segment

Central + Pharmacy is the single highest-profit region-channel combo at Rs. 21.4 Lac, providing a proven blueprint for underperforming combinations.

06 | Strategic Business Recommendations

R1 | Implement Discount Guardrails

Cap product-level discounts at 12% for top-10 revenue products. Night Repair Cream and BB Cream are currently discounted at 17%+ despite being best-sellers with strong organic demand. Reducing their discount to 12% could recover an estimated Rs. 30–40 Lac annually in net revenue — without impacting volume.

R2 | Fix the West Region Problem

West's 45.79% gross margin (vs. 60% in Central) demands immediate investigation. Likely causes: higher distributor discounts, unfavorable product mix, or aggressive competition. Deploying a Central-region Pharmacy approach in West could add Rs. 20–25 Lac to profit within one fiscal year.

R3 | Shift Sales Rep Incentives to Margin-Adjusted Revenue

Meena G generates the highest revenue but ranks 3rd in profit. A margin-adjusted incentive plan will reward reps like Vikram N who show better discount discipline and product-mix management, encouraging higher-quality sales behaviour across the team.

R4 | Scale the Central + Pharmacy Blueprint

The Central-Pharmacy combo is the single most profitable segment (Rs. 21.4 Lac). Replicate this success by aggressively expanding Pharmacy channel presence in the South and West regions, which currently underperform.

R5 | Protect Portfolio from Single-SKU Concentration Risk

Perfume 100ml EDP alone contributes 14% of total profit. Any supply disruption, pricing issue, or competitive entrant in this SKU could create a material impact. Invest in growing the next 3–5 products (Night Repair Cream, Under Eye Cream, Vitamin C Serum) to reduce this dependency.

R6 | Address the June Revenue Trough

June recorded the lowest monthly revenue at Rs. 51.7 Lac — Rs. 13.8 Lac below the peak month. A targeted mid-year campaign (Monsoon Beauty Essentials bundle, Salon Wholesale flash deals) could recover Rs. 8–10 Lac in June revenue.

07 | Analytical Concepts & Techniques Applied

Concept / Technique	Application
Data Ingestion	Loaded raw Excel/CSV into pandas DataFrame for structured processing
Data Cleaning	Null value removal via dropna(), duplicate detection, dtype correction
Date Feature Engineering	Extracted Year, Month, Quarter from order_date using .dt accessor
Financial Feature Engineering	Derived Gross Revenue, Discount Amount, Net Revenue, COGS, Gross Profit, Gross Margin %
ETL Pipeline	End-to-end Extract (Excel) → Transform (pandas) → Load (MySQL via SQLAlchemy)
SQL Aggregations	GROUP BY, SUM(), AVG(), COUNT(), ROUND() for business metric computation
SQL Ranking	ORDER BY + LIMIT for Top-N product, rep, and segment analysis
SQL Cross-tab Analysis	Multi-dimension GROUP BY (region x channel) for segment profitability
Margin Analysis	Gross Margin % as KPI for true profitability vs. revenue
Discount Impact Analysis	Quantifying revenue leakage from promotional discounting
Portfolio Concentration Analysis	Identifying SKU-level profit dependency risk
Comparative Performance Analysis	Revenue rank vs. profit rank for sales rep evaluation
Temporal Analysis	Month-on-month and quarter-on-quarter revenue trend identification
Power BI Visualization	Interactive .pbix dashboards for executive-ready reporting